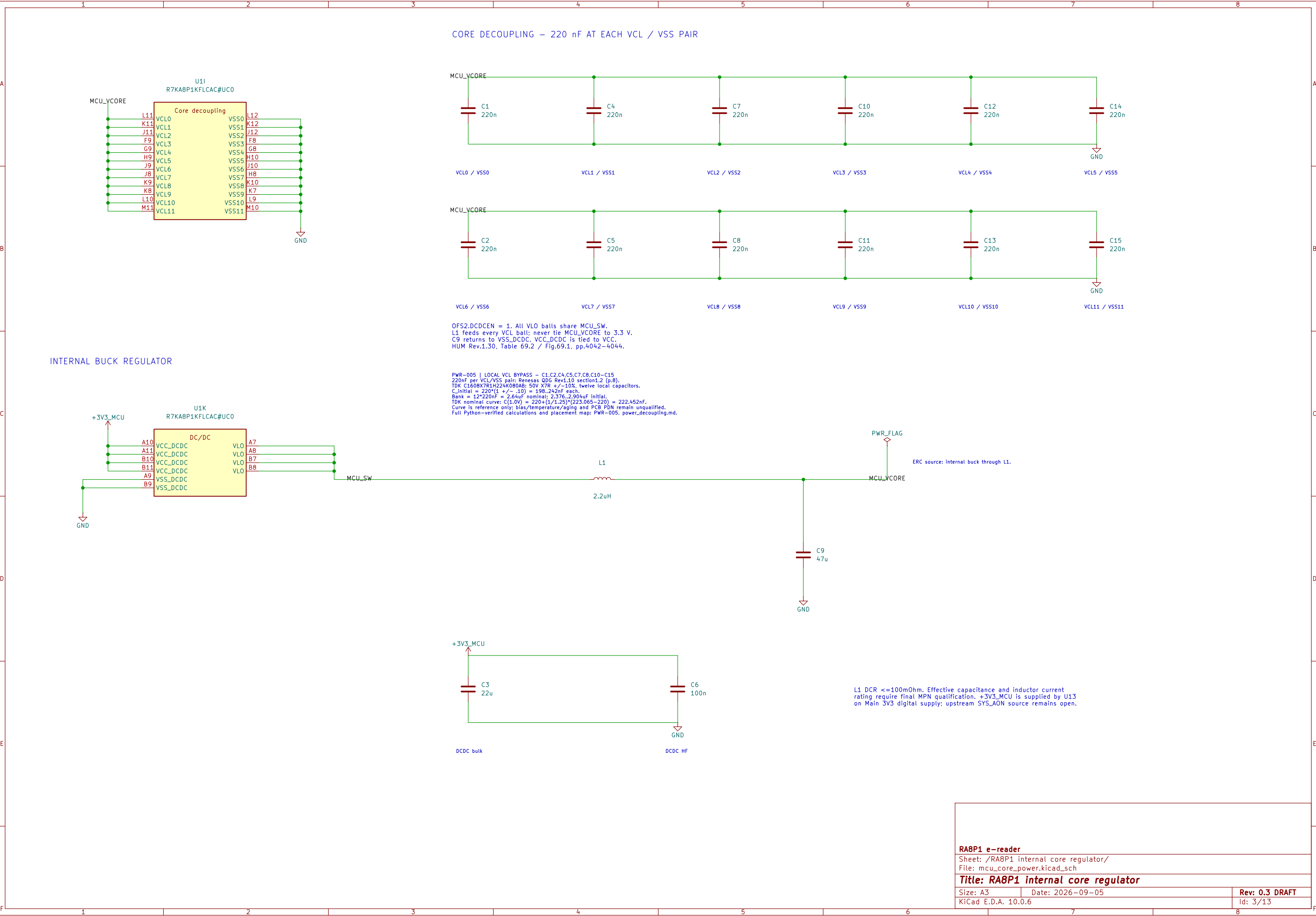
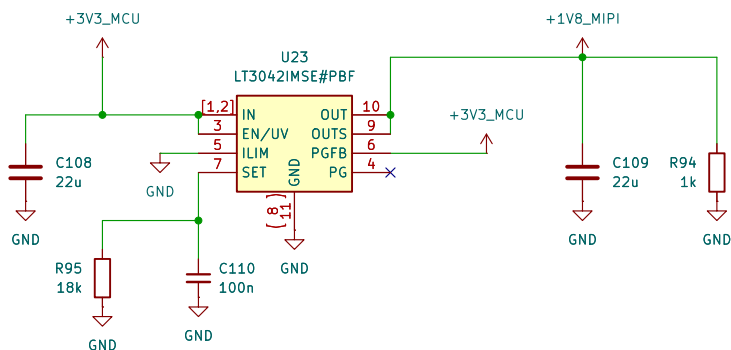


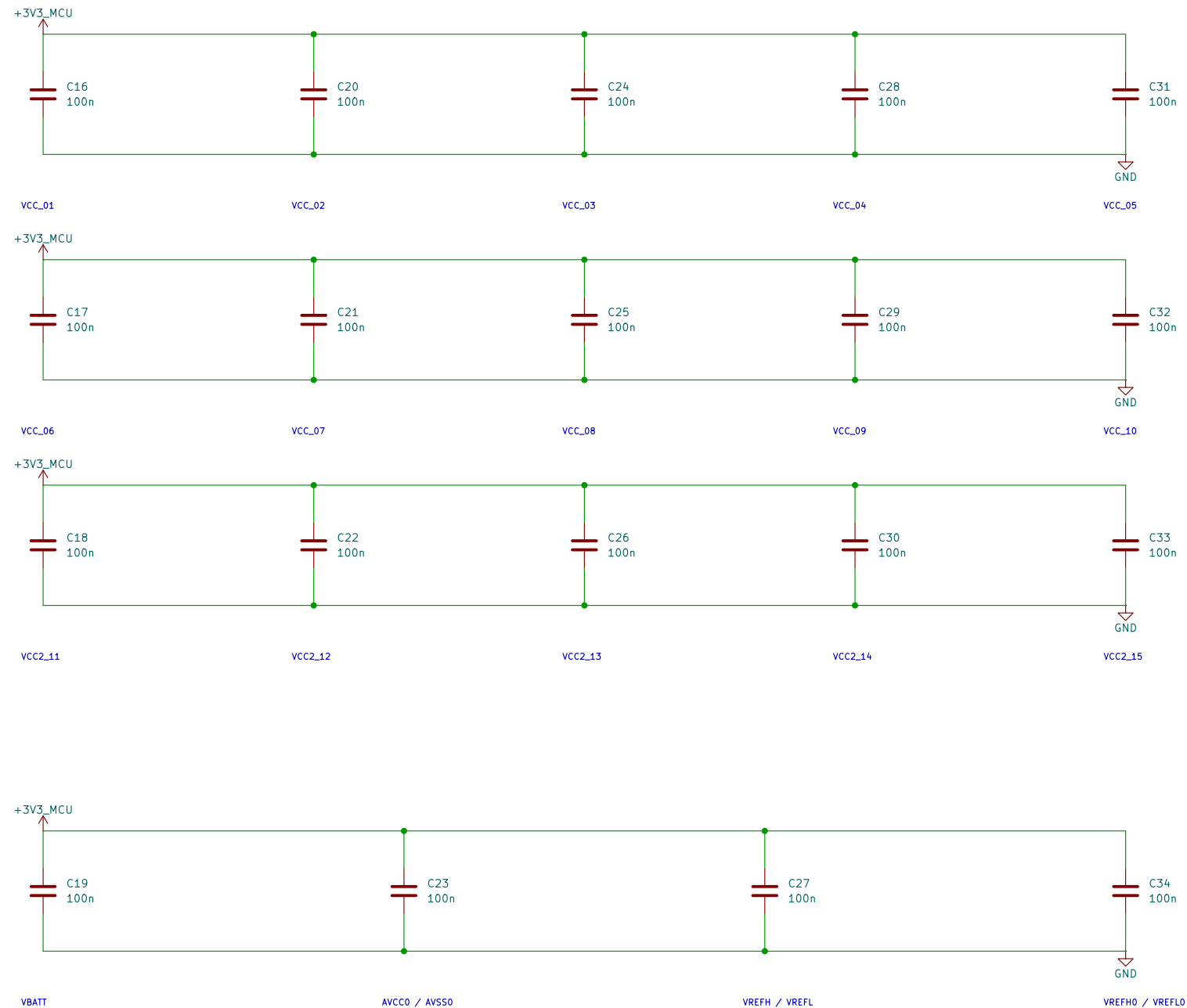
Development draft – not for fabrication.
Requirements: design/ereader_requirements.md
Tracking: GitHub epic #821.





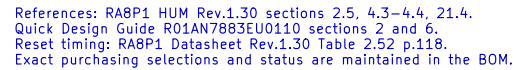
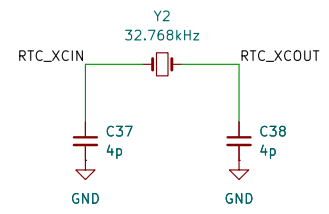
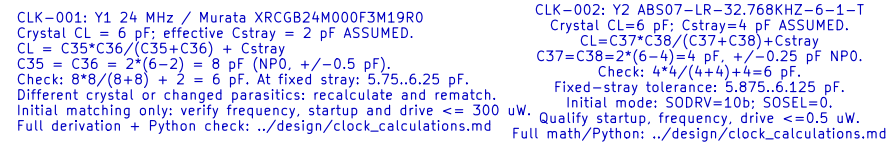


I/O DECOUPLING – 100 nF AT EACH NUMBERED VCC / VSS PAIR

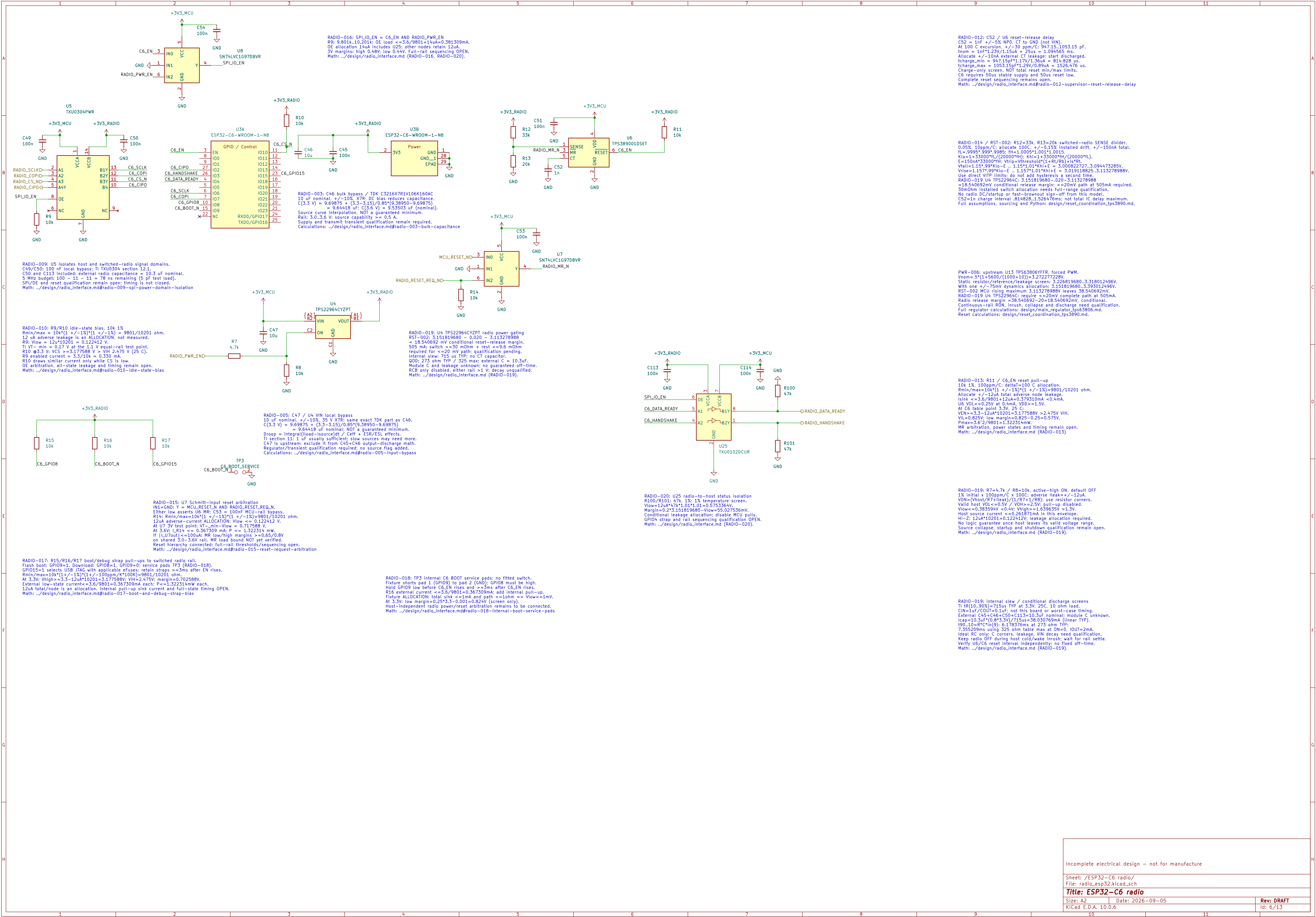


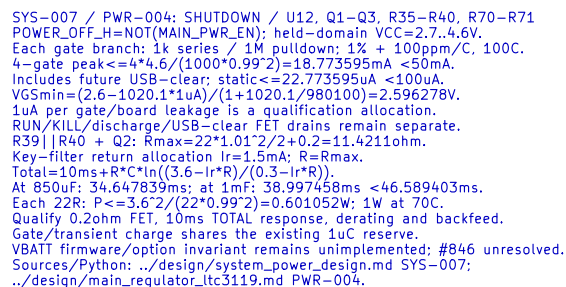
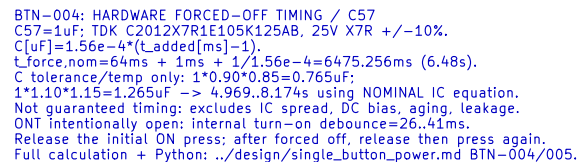
CMS-013: +20mA steady CSI branch planning allocation
 4.1mA PHY + 1.989593mA bleed + 13mA allocated GND
 + 0.102mA SET + 0.100mA auxiliary = 19.291593mA.
 13mA is an engineering envelope, not an interpolated guarantee.
 Auxiliary allowance requires verification: startup/inrush excluded.
 AVCC/MPI remains within existing MCU allocation.
 Main subunit: 2.075 + 0.020 = 2.095A; full system budget pending.
 At 2.0496V, 3.2V input, 1.25W average, 75% efficiency:
 input 2.961817158A; converter loss = 2.3694535727W.
 Source, current and thermal acceptance remain open.
 Sources, conditions and Python: design/camera_storage/interfaces.md.

RA8P1 e-reader		
Sheet: /RA8P1 IO and analog supplies/ File: mcu_supply_power.kicad_sch		
Title: RA8P1 I/O and analog supplies		
Size: A3	Date: 2026-09-05	Rev: 0.3 DRAFT
KiCad E.D.A. 10.0.6		Id: 4/13

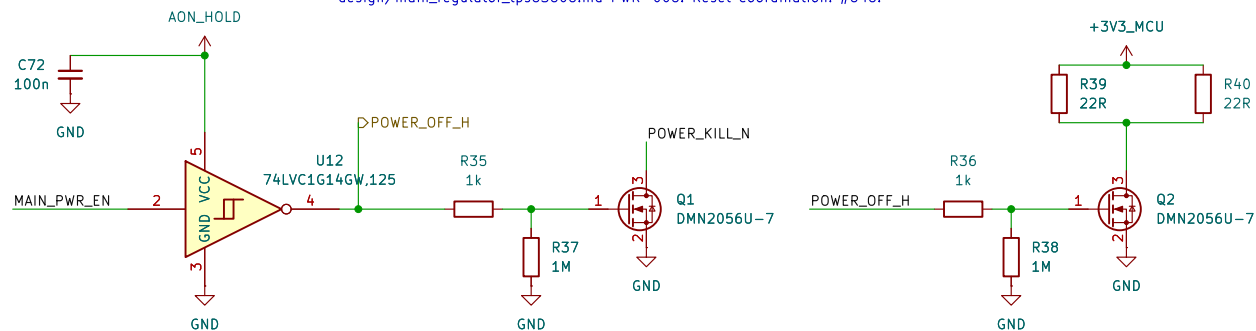
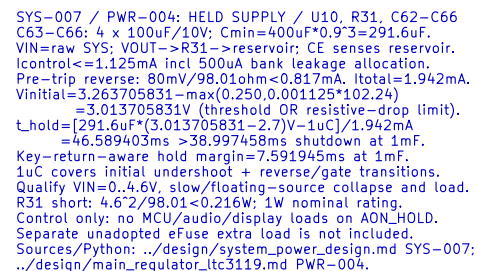


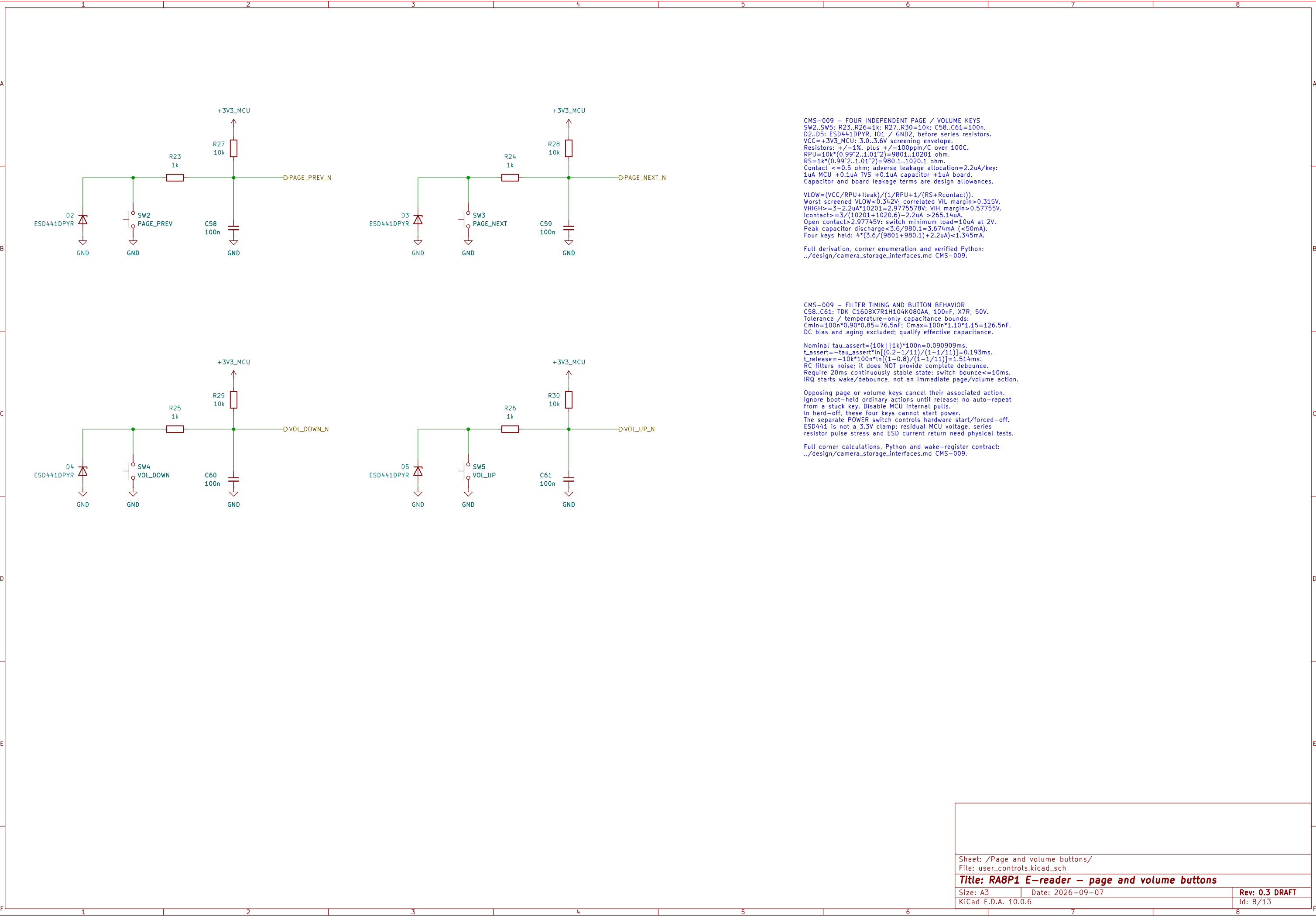
Sheet: /RA8P1 clocks, reset and debug/ File: mcu_clocks_debug.kicad_sch		
Title: RA8P1 clocks, reset and debug		
Size: A3	Date: 2026-09-05	Rev: 0.3 DRAFT
KiCad E.D.A. 10.0.6	Id: 5/13	





ERC supply path: U10 VOUT through R31.
PWR_FLAG declares that series-fed source only;
raw SYS_AON source remains unimplemented.





CMS-009 - FOUR INDEPENDENT PAGE / VOLUME KEYS
SW2..SW5; R23..R26=1k; R27..R30=10k; C58..C61=100n.
D2..D5: ESD441DPYR, IO1 / GND2, before series resistors.
VCC=+3V3_MCU; 3.0..3.6V screening envelope.
Resistors: +/-1%, plus +/-100ppm/C over 100C.
RPU=10k*(0.99^2..1.01^2)=9801..10201 ohm.
RS=1k*(0.99^2..1.01^2)=9801..10201 ohm.
Contact <=0.5 ohm; adverse leakage allocation=2.2uA/key:
1uA MCU +0.1uA TVS +0.1uA capacitor +1uA board.
Capacitor and board leakage terms are design allowances.

VLOW=(VCC/RPU+Ileak)/(1/RPU+1/(RS+Rcontact)).
Worst screened VLOW<0.342V; correlated VIL margin>0.315V.
VHIGH>=3-2.2uA*10201=2.9775578V; VIH margin>0.57755V.
Icontact>=3/(10201+1020.6)-2.2uA >265.14uA.
Open contact>2.97745V; switch minimum load=10uA at 2V.
Peak capacitor discharge<3.6/980.1=3.674mA (<50mA).
Four keys held: 4*(3.6/(9801+980.1)+2.2uA)<1.345mA.

Full derivation, corner enumeration and verified Python:
../design/camera_storage_interfaces.md CMS-009.

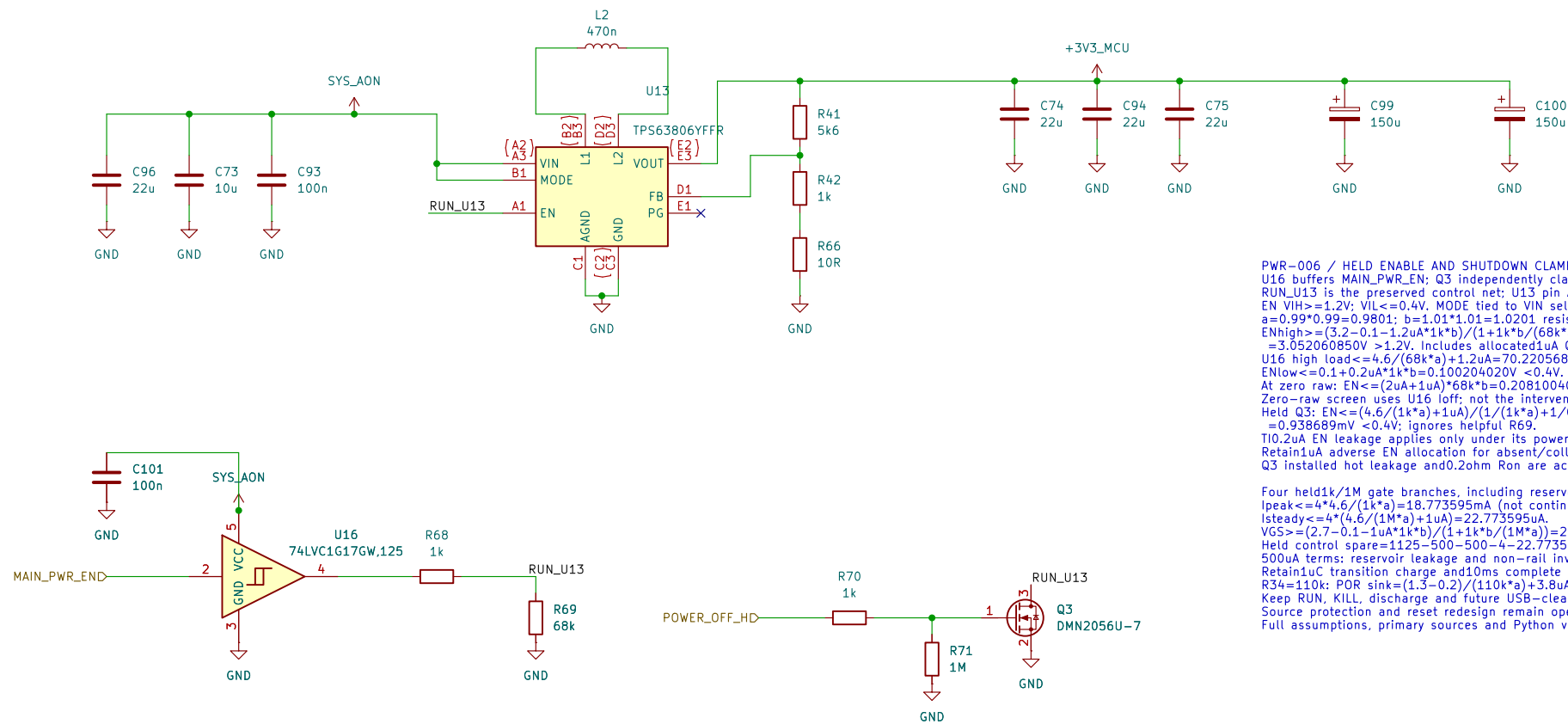
CMS-009 - FILTER TIMING AND BUTTON BEHAVIOR
C58..C61: TDK C1608X7R1H104K080AA, 100nF, X7R, 50V.
Tolerance / temperature-only capacitance bounds:
Cmin=100n*0.90*0.85=76.5nF; Cmax=100n*1.10*1.15=126.5nF.
DC bias and aging excluded; qualify effective capacitance.

Nominal tau_assert=(10k||1k)*100n=0.090909ms.
t_assert=-tau_assert*ln[(0.2-1/11)/(1-1/11)]=0.193ms.
t_release=-10k*100n*ln[(1-0.8)/(1-1/11)]=1.514ms.
RC filters noise; it does NOT provide complete debounce.
Require 20ms continuously stable state; switch bounce<=10ms.
IRQ starts wake/debounce, not an immediate page/volume action.

Opposing page or volume keys cancel their associated action.
Ignore boot-held ordinary actions until release; no auto-repeat
from a stuck key. Disable MCU internal pulls.
In hard-off, these four keys cannot start power.
The separate POWER switch controls hardware start/forced-off.
ESD441 is not a 3.3V clamp; residual MCU voltage, series
resistor pulse stress and ESD current return need physical tests.

Full corner calculations, Python and wake-register contract:
../design/camera_storage_interfaces.md CMS-009.

MAIN DIGITAL SUPPLY: TPS63806 / PWR-006
3.272277 V nominal; forced PWM; held shutdown clamp.
Divider and bypass calculations: design/main_regulator_tps63806.md
Reset coordination (#846), source protection and qualification remain open.
DEVELOPMENT SCHEMATIC – NOT FOR POWER-UP OR FABRICATION.



```

PWR-006 / HLD ENABLE AND SHUTDOWN CLAMP
U16 buffers MAIN.PWR.EN; Q3 independently clamps EN during raw collapse.
RUN_U13 is the preserved control net; U13 pin A1 is now EN.
EN VIH>=1.2V; VIL<=0.4V. MODE tied to V1N selects forced PWM.
a=0.99*0.99+0.9801; b=1.01*1.01-1.0201 resistor factors.
ENhigh>=(3.2-0.1-1.2uA*1k*b)/(1+1k*b*(68k*a))
=3.052060850V >1.2V. Includes allocated uA Q3 off leakage.
U16 high load <=4.6/(68k*a)+1.2uA=70.220568uA <100uA.
ENlow<=0.1+0.2uA*1k*b+0.100204020V <0.4V.
At zero raw: EN<=(2uA+1uA)*68k*b=0.208100400V.
Zero-raw screen uses U16 loff; not the intervening undefined region.
Held Q3: EN<=(4.6/(1k*a)+1uA)/(1/(1k*a)+1/0.2ohm)
=0.938689mV <0.4V; ignores helpful R69.
T10.2uA EN leakage applies only under its powered test conditions.
Retain1uA adverse EN allocation for absent/collapsing raw.
Q3 installed hot leakage and0.2ohm Rn are acceptance allocations.

```

```

Four 4h1dk/1M gate branches, including reserved USB-clear:
Ipeak<=4*4.6/(1k* $\mu$ )=18.773595mA (not continuous rating).
Isteady<=4*4.6/(1M* $\mu$ +1 $\mu$ )=22.773595uA.
VGS>=(2.7-0.1-1uA/1k* $\mu$ )/(1+1k* $\mu$ /(1M))=2.596278V.
Held current spare=1125-500-500-4-22.773595=98.226405uA.
500uA terms: reservoir leakage and non-rail inverter-injection allocation.
R34=10k: transition resistance and gate leakage.
R34=110k: POR sink=(1.3-0.2)/(110k* $\mu$ +3.8uA)=14.003041uA.
Keep RUN, KILL, discharge and future USB-clear drains separate.
Source protection and reset redesign remain open; no power-up approval.
Full assumptions, primary sources and Python verification: click this note.

```

PWR-Vom / FEEDBACK AND LOCAL PASSIVES
 $V_{nom} = 5.00 \times (1 + 5600 / (1000 + 10)) = 3.27227228 \text{ V}$
R41: 0.02%, 5ppm/C; R42: 0.01%, 2ppm/C; R66: 1%, 200ppm/C.
Installed boards: initial/TRC(1000) "aging" assembly, then ohms.
 $R41 = 5600 \times (1 + (-0.0002)) \times (1 + (-0.0005)) \times (1 + (-0.001))$
 $\times (1 + (-0.0005)) \div (-0.02) = 5587.6629237, 5612.349243 \text{ ohm}$
 $R42 = 1000 \times (1 + (-0.0001)) \times (1 + (-0.0002)) \times (1 + (-0.001))$
 $\times (1 + (-0.0005)) \div (-0.01) = 998.190970, 1001.810970 \text{ ohm}$
 $R66 = 10 \times (1 + (-0.01)) \times (1 + (-0.02)) \times (1 + (-0.01)) \div (-0.05)$
 $= 9.954980, 10.455020 \text{ ohm}$ one combined aging/assembly 1%.
 $V_{in} = 5.00 \times (1 + 141680 / (R41 \times R42 \times R66)) = 1.000001 \times 1.000001 \times 1.000001$
 $\times V_{in} = 5.005 \times (1 + R41 \times R42 \times R66) = 5.005 \times (1 + 100nA \times R41 \times \text{max})$
 $V_{static} = 3.226819680, 3.318012496 \text{ V (FPWM only)}$
One combined + / -75mV regulation/ripple/transient/routing allowance:
 $V_{complete} = 3.151819680, 3.393012496 \text{ V}$: qualification required.
SDRAM static VOH margin = $2.4 - 0.73 = 3.93012496 \text{ V} = 24.891253 \text{ mV}$.
This is not measured S/I/noise overhead or reset qualification.

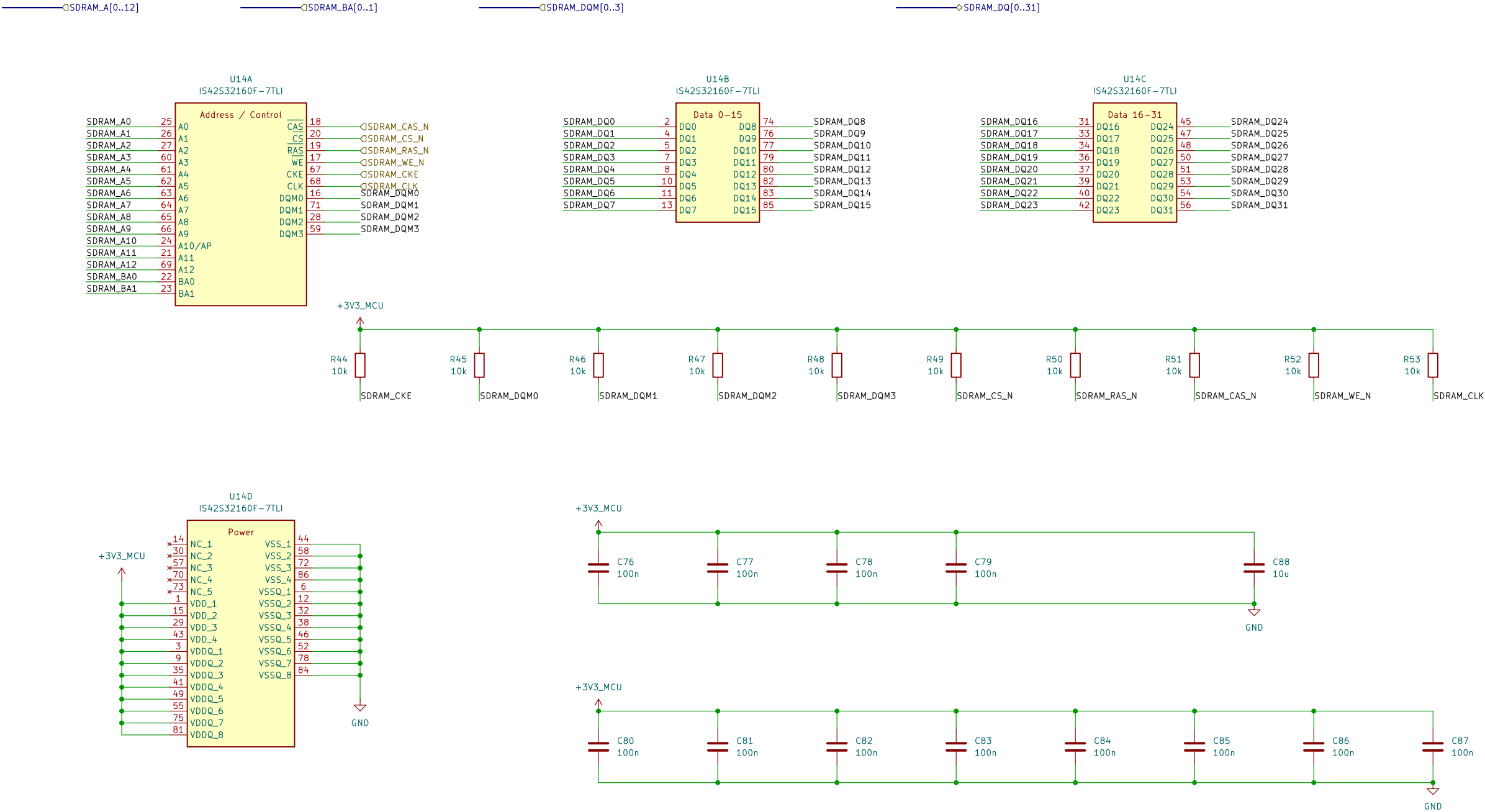
12: 470nH +/-20%; catalog spread alone is insufficient.
Screen incoming/0.412-0.518uH; allocated operating +/-10%
=>0.3708-0.5698uH versus 0.10.87-0.57uH. Quality hot/current.
C74/C75/C94: $3.22uF \pm 0.85 \cdot 0.6 = 2.628uF \pm 0.121uF$.
C96: $22uF \pm 0.8 \cdot 0.85 \cdot 0.6 = 8.97uF \pm 0.14uF$ input minimum.
Circuit board combined input capacitance not intended.
Sampling typical output-bias retention at3.393012496V=90.221554%.
Input bias, aging, mounted impedance and startup need verification.
Full assumptions, sources and executable Python: click this note.

PWR-006 / LDA, STORAGE AND SHUTDOWN
Allocation: 1.80A operating; 2.25A cold start screen, not measured load.
Pwr: 3.393012496e+1, 8=6.107424332
Thermal: 2.544759372A, 2=2.544759372A
Losses: 0.67860299W, 2=0.67860299W
TJ90C efficiency: loss=0.67860299W.
TJ90C=70C+78.8C/0.67860299W=123.474C.
2.544759372A is not the final PCB thermal resistance.
TJ90C=70C+78.8C/3.393012496e+1=120.179037499W.
Converter limit, source limit, thermal and thermal envelope need validation.

Native storage inventory (nominal, not effective minimum):
 Direct main469.21uF + C43 through FB1 10uF + key filters0.4uF
 =479.61uF cold; radio-on add±0.2uF =>489.81uF.
 C99/C100 remain±150uF each. C94 adds22uF to the output.
 Old bootstrap/private VCC/VF capacitors were not main-rail storage.
 Whole-main ceiling remains1mF for shutdown; no LTC stability transfer.
 Cout lower acceptance and distributed PDN/startup are separate checks.

Preserved discharge screen: Rmax11.4211ohm; key return<=1.5mA.
Vinf=1.5mA*11.4211ohm=0.01713165V.
t=1mF*11.4211ohm*n((3.6-Vinf)/(0.3-Vinf))+10ms
=38.997458ms to 0.3V. (b6 held 46.589403ms screen.
Response10ms includes command/clamp/discharge delays.
Held budget retains1uC transition charge and98.226405uA spare.
SYS_AON source protection, loaded start and fast faults remain open.
RST-002 / RADIO-019: release margin remains conditional on rail/path bounds.
Do not power up until reset coordination and source protection are complete.
Full assumptions, sources and executable Python: click this note.

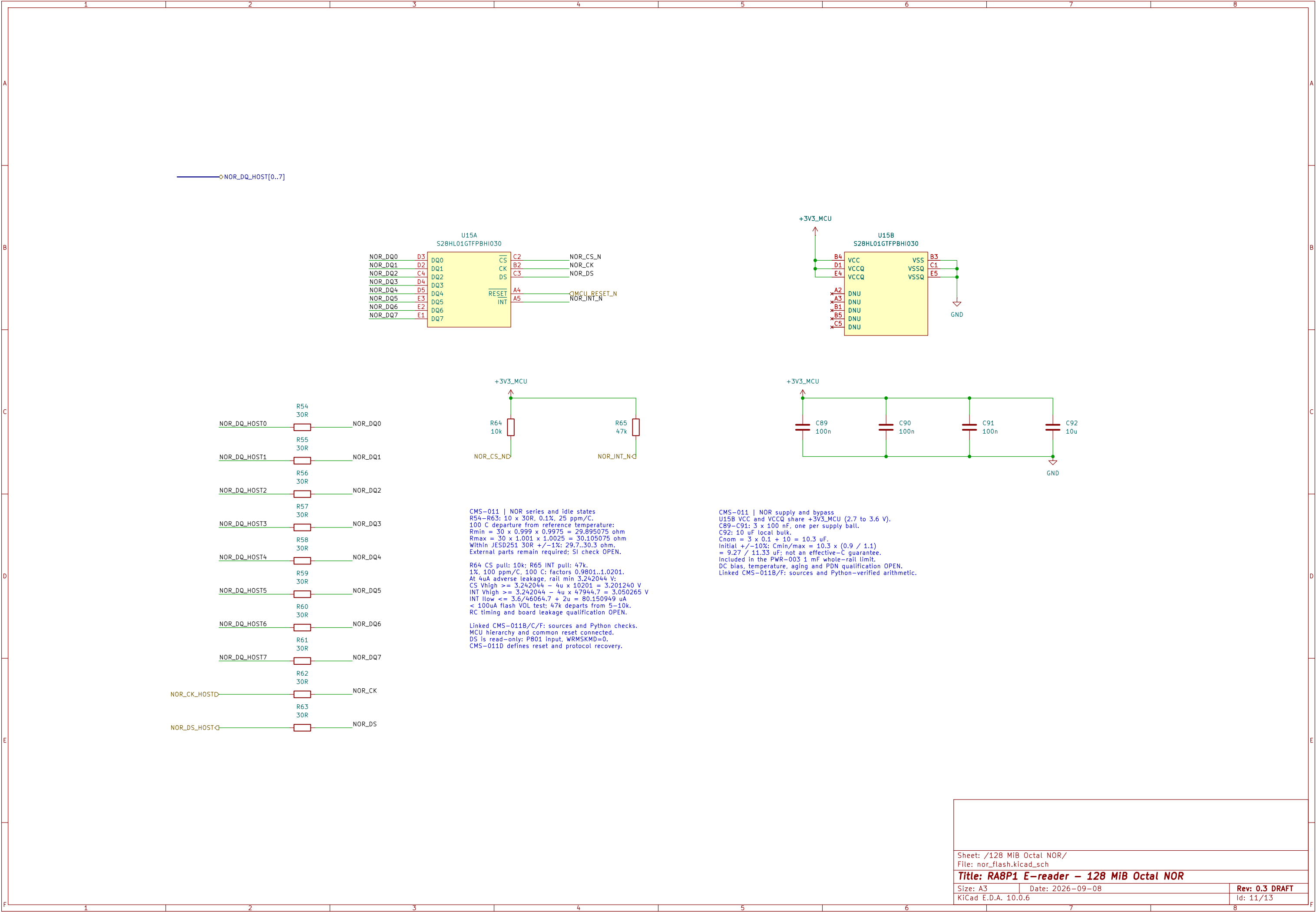
64 MiB SDRAM | ISSI IS42S32160F-7TLI
Signal and reset—default circuits | CMS-010.
Startup, SI/timing and hardware qualification remain open.



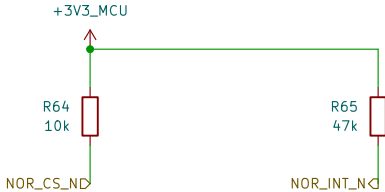
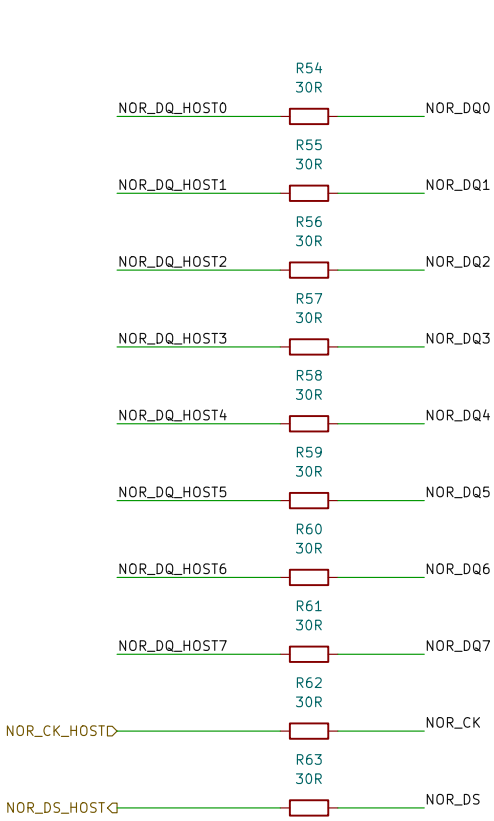
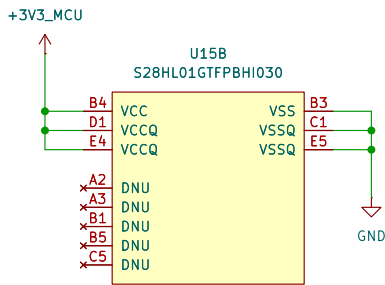
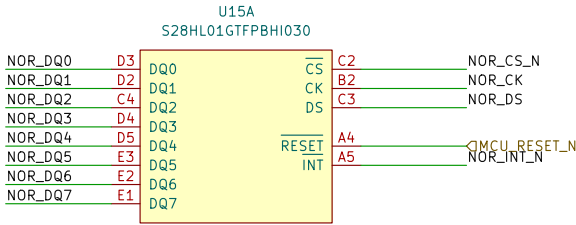
CMS-010 | SDRAM power and bypass
VDD and VDDQ share +3V3_MCU (3.0 to 3.6 V).
C76-C79: 4 x 100 nF for the four VDD pins.
C80-C87: 8 x 100 nF for the eight VDDQ pins.
C88: 10 uF local bulk; Cnom = 12 x 0.1 + 10 = 11.2 uF.
Included in the PWR-003 1 mF whole-rail limit.
Effective C, placement and PDN impedance require qualification.

CMS-010 | R44-R53: 10k reset defaults
Rail 3.0..3.6V: +/-1%, 100ppm/C. |dT|<=100C.
Rmin=10k*0.99*0.99=9801 ohm.
Rmax=10k*1.01*1.01=10201 ohm.
Ioff=5uA SDRAM+1uA MCU+1uA board=7uA.
VHIGHmin=3.0-7uA*10201=2.928593V (>2.0V VIH).
Board leakage and temperature screen require qualification.

CMS-010 | Pull load and power states
Sink leakage: 5uA SDRAM+1uA board=6uA.
Isink=3.6/9801+6uA=0.373309mA (<0.374mA).
Pmax=3.6^2/9801=1.322314mW per pull.
All ten LOW: 10*3.6/9801=3.673095mA rail:
10*3.6^2/9801=13.223140mW resistor heat.
Reserve 4mA within the 250mA SDRAM allocation.
CKE/DQM and CS_N default HIGH; CLK HIGH is not a clock.
Switched-rail pulls; self-refresh actively holds CKE LOW.
Startup, dynamic I/O, SI and shutdown qualification OPEN.



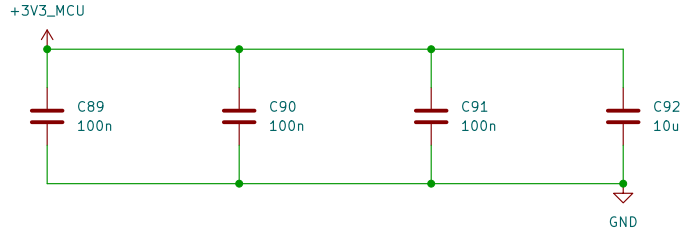
NOR_DQ_HOST[0..7]



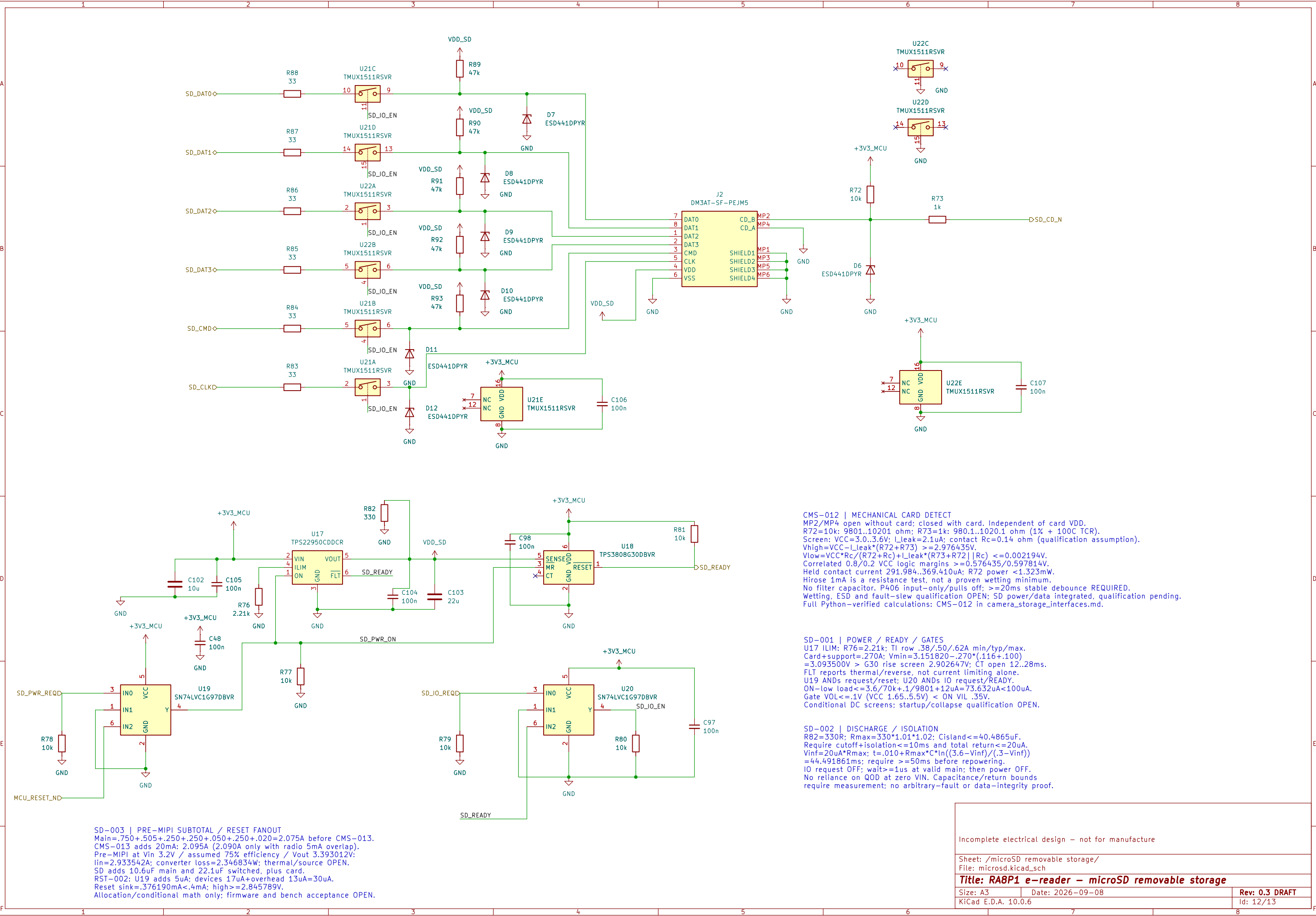
CMS-011 | NOR series and idle states
R54-R63: 10 x 30R, 0.1%, 25 ppm/C.
100 C departure from reference temperature:
 $R_{min} = 30 \times 0.999 \times 0.9975 = 29.895075 \text{ ohm}$
 $R_{max} = 30 \times 1.001 \times 1.0025 = 30.105075 \text{ ohm}$
Within JESD251 30R +/-1%; 29.7..30.3 ohm.
External parts remain required; SI check OPEN.

R64 CS pull: 10k; R65 INT pull: 47k.
1%, 100 ppm/C, 100 C: factors 0.9801..1.0201.
At 4uA adverse leakage, rail min 3.242044 V:
 $CS_{Vhigh} \geq 3.242044 - 4u \times 10201 = 3.201240 \text{ V}$
 $INT_{Vhigh} \geq 3.242044 - 4u \times 47944.7 = 3.050265 \text{ V}$
 $INT_{Ilow} \leq 3.6/46064.7 + 2u = 80.150949 \text{ uA}$
< 100uA flash VOL test: 47k departs from 5-10k.
RC timing and board leakage qualification OPEN.

Linked CMS-011B/C/F: sources and Python checks.
MCU hierarchy and common reset connected.
DS is read-only; P801 input, WRMSKMD=0.
CMS-011D defines reset and protocol recovery.



CMS-011 | NOR supply and bypass
U15B VCC and VCCQ share +3V3_MCU (2.7 to 3.6 V).
C89-C91: 3 x 100 nF, one per supply ball.
C92: 10 uF local bulk.
 $C_{nom} = 3 \times 0.1 + 10 = 10.3 \text{ uF}$
Initial +/-10%; $C_{min}/max = 10.3 \times (0.9 / 1.1)$
 $= 9.27 / 11.33 \text{ uF}$: not an effective-C guarantee.
Included in the PWR-003 1 mF whole-rail limit.
DC bias, temperature, aging and PDN qualification OPEN.
Linked CMS-011B/F: sources and Python-verified arithmetic.



CMS-012 | MECHANICAL CARD DETECT
MP2/MP4 open without card; closed with card. Independent of card VDD.
R72=10k; 9801..10201 ohm; R73=1k; 9801..10201 ohm (1% + 100C TCR).
Screen: VCC=3.0..3.6V; I_{Leak}=2.1uA; contact R_c=0.14 ohm (qualification assumption).
V_{high}=VCC-I_{Leak}*(R72+R73) >=2.976435V.
V_{low}=VCC*R_c/(R72+R_c)+I_{Leak}*(R73+R72)|R_c <=0.002194V.
Correlated 0.8/0.2 VCC logic margins >=0.576435/0.597814V.
Held contact current 291.984..369.410uA; R72 power <1.323mW.
Hirosi 1mA is a resistance test, not a proven wetting minimum.
No filter capacitor. P406 input-only/pulls off; >=20ms stable debounce REQUIRED.
Wetting, ESD and fault-slew qualification OPEN; SD power/data integrated, qualification pending.
Full Python-verified calculations: CMS-012 in camera_storage_interfaces.md.

SD-001 | POWER / READY / GATES
U17 ILIM: R76=2.21k; TI row .38/.50/.62A min/typ/max.
Card+support=.270A; V_{min}=3.151820-.270*(.116+.100)
=3.093500V > G30 rise screen 2.902647V; CT open 12..28ms.
FLT reports thermal/reverse, not current limiting alone.
U19 ANDs request/reset; U20 ANDs IO request/READY.
ON-low load<=3.6/70k+.1/9801+12uA=73.632uA<100uA.
Gate VOL<=.1V (VCC 1.65..5.5V) < 0N VIL .35V.
Conditional DC screens; startup/collapse qualification OPEN.

SD-002 | DISCHARGE / ISOLATION
R82=330R; R_{max}=330*1.01*1.02; Cisland<=40.4865uF.
Require cutoff+isolation<=10ms and total return<=20uA.
V_{inf}=20uA*R_{max}; t=.010+R_{max}*C*ln((3.6-V_{inf})/(3-V_{inf}))
=44.491861ms; require >=50ms before repowering.
IO request OFF; wait>=1us at valid main; then power OFF.
No reliance on QOD at zero VIN. Capacitance/return bounds
require measurement; no arbitrary-fault or data-integrity proof.

SD-003 | PRE-MIPI SUBTOTAL / RESET FANOUT
Main=.750+.505+.250+.250+.050+.250+.020=2.075A before CMS-013.
CMS-013 adds 20mA: 2.095A (2.090A only with radio 5mA overlap).
Pre-MIPI at Vin 3.2V / assumed 75% efficiency / Vout 3.393012V:
I_{in}=2.933542A; converter loss=2.346834W; thermal/source OPEN.
SD adds 10.6uF main and 22.1uF switched, plus card.
RST-002: U19 adds 5uA; devices 17uA+overhead 13uA=30uA.
Reset sink=.376190mA<.4mA; high>=2.845789V.
Allocation/conditional math only; firmware and bench acceptance OPEN.

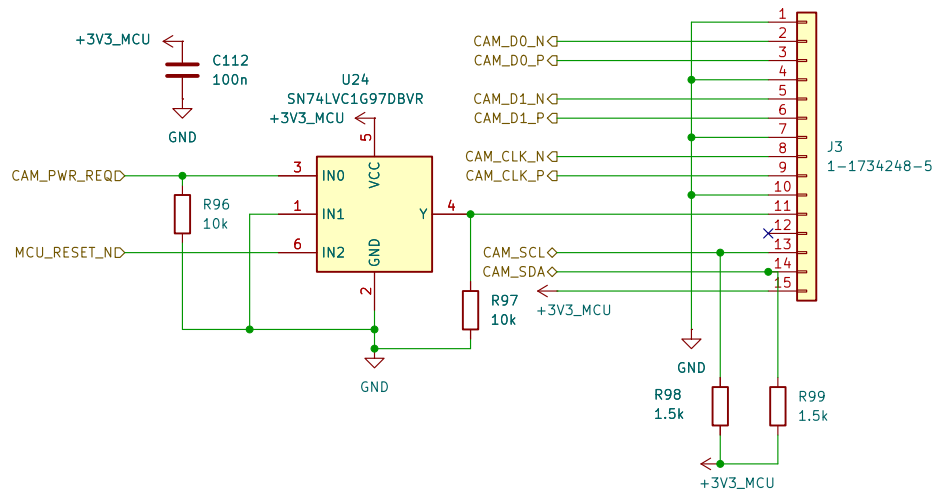
Incomplete electrical design – not for manufacture

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File: microsd.kicad_sch

Title: RA8P1 e-reader – microSD removable storage

Size: A3 Date: 2026-09-08
KiCad E.D.A. 10.0.6

Rev: 0.3 DRAFT
Id: 12/13



Sheet: /Pcam 5C camera/
File: camera.kicad_sch

Title:

Size: A4 Date:

KiCad E.D.A. 10.0.6

Rev:

Id: 13/13